Effective Spatial–Spectral Feature Representation for Hyperspectral Image Classification

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***Abstract*—Hyperspectral images (HSIs), with their rich spectral information and spatial details, have demonstrated significant potential for classification tasks in fields such as remote sensing, agriculture, and environmental monitoring. However, existing methods still exhibit limitations in feature representation, primarily manifested in insufficient contextual modeling and the inability to effectively address spectral redundancy and significant variations in spatial scales. To address these challenges, this article proposes an enhanced MambaHSI-based framework for HSI classification, focusing on improving the representation capability of spatial–spectral features. The proposed method consists of three key innovations: 1) a hierarchical DualGroupMamba module that progressively models intragroup and intergroup spectral dependencies to enhance fine-grained spectral discrimination and global contextual awareness; 2) a lightweight hyperspectral channel attention (HCA) that dynamically adjusts the importance of feature channels based on the spatial–spectral information of different bands, effectively suppressing redundant information and highlighting discriminative features; and 3) a hybrid feature enhancer (HFE) module that effectively represents and fuses multiscale spatial features by extracting local texture details and perceiving the overall spatial distribution of scenes, thereby enhancing the model's adaptability to complex spatial structures. Through a systematic evaluation on four benchmark hyperspectral datasets, the proposed method achieved an average overall classification accuracy of 95.64%, outperforming the current best method by 1.89%. The experimental results validate the superior performance of the proposed approach in enhancing the representation of spatial–spectral features. The latest logs are now available at https://github.com/Tomyaya/EFR**

***Index Terms*—Dynamic channel adjustment, hyperspectral image (HSI) classification, Mamba architecture, multiscale spatial features, remote sensing, spatial–spectral feature representation.**

I. Introduction

ONGOING progress in Earth observation has facilitated broader access to hyperspectral imagery [1]. Compared to RGB images with only three channels, hyperspectral images (HSIs) typically consist of dozens to hundreds of narrow spectral bands, enabling the capture of subtle spectral variations of materials across different wavelengths. These rich spectral characteristics, combined with spatial information, facilitate more comprehensive and detailed feature analysis, enabling HSIs to be widely applied in classification tasks across fields such as remote sensing, agriculture, and environmental monitoring [2], [3], [4], [5], [6].

To address the problem of HSI classification, researchers have proposed a variety of methods. Traditional learning methods include support vector machines (SVMs) [7], random forests (RFs) [8], and k-nearest neighbors (KNNs) [9], [10], [11]. However, they struggle to fully exploit the spatial information embedded in hyperspectral data, which limits the performance of such methods [12]. Meanwhile, in deep learning (DL) methods, convolutional neural networks (CNNs) struggle to effectively model long-range dependencies due to their fixed receptive field [13], [14], [15]. The Transformer architecture, with its self-attention mechanism, offers a more flexible and efficient solution for modeling long-range dependencies [16], [17]. However, its computational complexity grows quadratically with sequence length ($\mathcal{O}(N^2)$), resulting in substantial computational overhead when processing hyperspectral data containing hundreds of spectral bands [18], [19], [20], [21].

Mamba, built upon the state-space model (SSM), provides an effective solution for HSI classification by fully aggregating contextual information while preserving a low linear computational complexity of $\mathcal{O}(N)$. For example, SpectralMamba [22] and SS-Mamba [23] focus on the fusion of spatial and spectral features. S2Mamba [24] and WaveMamba [25] emphasize multiscale information modeling. Meanwhile, Mamba-in-Mamba (MiM) [26] and IGroupSS-Mamba [27] further enhance representational capacity through module integration or hierarchical design. DBMGNet [28] combines graph convolutional networks (GCNs) and Mamba for joint spatial–spectral modeling. However, existing methods often overlook the intrinsic correlation between adjacent spectral bands, making it difficult to effectively capture representative spectral features. For example, neighboring bands usually exhibit strong correlation, which can be leveraged to design more effective spectral grouping strategies. MambaHSI [29] introduces a grouped approach by dividing the long spectral sequence of each pixel into multiple locally continuous groups. This grouping strategy enables the model to capture overall differences across groups within the spectrum, thereby facilitating the extraction of more discriminative spectral features for classification. Nevertheless, this approach fails to explicitly model intragroup spectral contexts, neglecting fine-grained

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variations within local spectral regions, thereby limiting its performance.

Moreover, in terms of feature representation, most existing Mamba-based improved methods predominantly focus on optimizing contextual modeling and feature fusion strategies, yet lack specialized designs tailored to the inherent spatial–spectral characteristics of HSIs, thereby further limiting the performance of current methods. Specifically, accurate classification fundamentally relies on discriminative key features rather than consistent or approximate representations. The spatial scale of HSIs is large, and each pixel encapsulates rich spectral information. While this enhances the potential of hyperspectral imagery for classification tasks, it may simultaneously introduce approximate noise, thereby hindering the effective capture and extraction of critical discriminative features. Therefore, it is imperative to further select representative features from the massive spatial–spectral information to enhance classification performance. Additionally, hyperspectral imagery typically exhibits complex geometric boundary configurations and nonuniform background interference noise. This requires classification models not only to accurately capture local details but also to effectively perceive multiscale spatial structures, thereby imposing higher demands on multiscale spatial modeling.

To address the aforementioned challenges, in this article, we propose a novel HSI classification method based on an improved MambaHSI framework, which includes three key modules. First, to address the issue of insufficiently fine spectral modeling, we introduce a novel hierarchical dual-stage spectral modeling module—DualGroupMamba. Based on spectral grouping, this module first models the intragroup dependencies among spectral bands within each spectral subgroup using Mamba, and then captures the overall contextual information at the intergroup level. This progressive modeling strategy preserves sensitivity to local details while incorporating richer global semantic information, thereby enhancing the feature representation capability.

Compared with traditional CNN- or Transformer-based spectral modeling methods, DualGroupMamba enables finer-grained dependency extraction by decomposing the spectral sequence into local and global groups. Moreover, unlike existing Mamba-based methods only model the full spectrum as a whole, our structure enables the model to preserve fine-grained local details while capturing global contextual trends, significantly enhancing the spectral representation capability.

In response to the challenge of limited feature discriminability, we introduce the hyperspectral channel attention (HCA) module, which employs global average pooling (GAP) to extract channel attention information from spatial–spectral features, while adaptively assigning weights to different channels through a lightweight attention mechanism to filter more discriminative spectral features.

To further tackle the modeling complexity across spatial scales, we propose a hybrid feature enhancer (HFE) module to augment spatial information modeling capabilities. The HFE adopts a multibranch architecture that synergistically integrates local depthwise convolution with global pooling perception. By establishing correlations between local texture features and global scene features, it effectively improves the model's ability to extract and fuse multiscale spatial features.

Our contributions are as follows.

1) A novel DualGroupMamba module is proposed, which adopts a two-stage structure of "intragroup modeling—intergroup modeling" to conduct comprehensive contextual modeling of spectral information, thereby enhancing the characterization capability of spectral features.
2) A lightweight HCA module is constructed, which integrates channel attention mechanisms with residual connections, thereby emphasizing representative channel features and improving the discriminability of spatial–spectral features.
3) An HFE module is introduced, which employs a dual-branch fusion strategy integrating deep convolution and global pooling mechanisms, effectively enhancing the model's ability to capture and integrate multiscale spatial features.

We conducted extensive experiments on four real HSI datasets. The results demonstrate that the proposed method shows significant advantages in enhancing feature representation ability and outperforms current mainstream advanced HSI classification models based on the CNN, Transformer, and Mamba in terms of classification performance.

II. Related Work

HSI classification approaches primarily encompass traditional machine learning methods and recently emerging DL techniques. This section provides a comprehensive overview of these methodologies.

A. Traditional Machine Learning Approaches

Early hyperspectral classification methods largely relied on traditional machine learning algorithms such as SVMs [7] and RFs [8], which focus on classifying the spectral features of individual pixels. These methods suffer from poor performance due to their inability to capture the contextual information of neighboring pixels [3]. The KNNs method [9], [10], [11] can alleviate the issue of spatial information neglect in SVMs and RFs to some extent. However, it relies on distance metrics and cannot thus learn and model spatial context over a larger range. In addition, there are unsupervised feature learning frameworks based on multilayer dropout K-means networks for learning spatial features [30], as well as other unsupervised methods that combine K-means and principal component analysis (PCA) [31]. Nevertheless, these models are only able to extract shallow features when dealing with complex spectral characteristics of land cover, and thus their performance is generally unsatisfactory.

B. DL Approaches

DL, with its powerful feature learning capability, can extract more abstract and discriminative feature representations from HSI data. Main approaches fall into three categories: CNN-based, Transformer-based, and the emerging Mamba-based methods.

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1) CNN-Based: With the widespread adoption of CNNs [32], [33], [34], [35] and recurrent neural networks (RNNs) [36] in HSI classification, research has increasingly focused on spectral feature extraction, contextual modeling, and sequential pattern learning. Hybrid architectures such as convolutional RNNs (CRNNs) have been proposed to integrate spatial constraints into spectral modeling, further improving classification performance [37]. GCNs [38], [39], [40] are particularly effective in handling irregular input structures and can capture both local and global contextual relationships, offering a promising direction for advanced HSI classification. Although CNN-based models demonstrate excellent performance, their fixed input dimensions and limited receptive fields restrict the model's flexibility and its ability to capture complex spatial information.

2) Transformer-Based: Transformer-based architectures have been widely adopted in HSI classification due to their powerful self-attention mechanisms. These models significantly enhance the representational capacity of HSI data by effectively modeling long-range dependencies. Pioneering works first introduced Transformers to HSI classification by grouping adjacent spectral bands to jointly model local and global information [41]. Subsequent studies have developed various hybrid architectures: CNN–Transformer combinations that exploit convolutional operations for local feature extraction [42], [43], [44], RNN-enhanced Transformers that better model spectral sequentiality [45], and biologically inspired center-surround mechanisms for multiscale representation learning [46]. Additional refinements include specialized modules for local detail enhancement [47], [48], lightweight designs optimized for data-scarce scenarios [49]. Despite performance improvements, the quadratic complexity of self-attention remains a significant challenge for computational efficiency and memory consumption, ultimately limiting its scalability for modeling long-range dependencies in hyperspectral data. While Transformers have achieved great success in various vision tasks, this complexity restricts their practical application in high-dimensional hyperspectral scenarios. In contrast, Mamba's linear-time modeling approach offers greater computational efficiency without sacrificing the ability to capture contextual dependencies, making it well-suited for HSI classification.

C. Mamba-Based Approaches

Mamba is a novel sequence modeling approach based on continuous SSMs, which reformulates sequence modeling as a first-order recurrence, enabling global dependency modeling through linear dynamics [50]. Unlike Transformers that rely on fully connected attention mechanisms, Mamba adopts a structured state update process that avoids explicit interactions across all positions, resulting in linear computational complexity and improved efficiency for long sequences. In HSIs, each pixel contains hundreds of contiguous spectral bands, exhibiting strong local continuity and cross-channel dependencies. This makes HSIs naturally well-suited for continuous state modeling. Leveraging its advantages in modeling continuous signals, Mamba has been increasingly applied to HSI tasks such as spectral modeling, architectural design, and spatial–spectral fusion, demonstrating superior performance compared to traditional attention-based methods.

SpectralMamba [22] introduces a state-space mechanism along the spectral dimension to effectively model dependencies between spectral bands. WaveMamba [25] leverages multiscale wavelet decomposition to enhance sensitivity to spectral frequency heterogeneity. S2Mamba [24] adopts a dual-branch structure to achieve independent modeling of spatial and spectral dimensions. Building on this, IGroupSS-Mamba [27] proposes an iterative intragroup feature fusion mechanism to further strengthen spatial–spectral interaction. MiM [26] deepens feature modeling through a nested hierarchical structure, while SS-Mamba [23] integrates self-attention with the Mamba module to balance local perception and global modeling capabilities. DBMGNet [28] integrates the Mamba SSM with GCNs to simultaneously model long-range dependencies and similarity in non-Euclidean space, thereby enhancing classification performance. These methods expand the representational capacity of Mamba in HSI processing from various perspectives, significantly boosting model performance. Among them, MambaHSI [29] was the first to propose a global modeling approach across the entire image by unfolding the HSI into a 2-D sequence and designing targeted processing strategies in both the spatial and spectral domains, thereby enabling more effective feature extraction and fusion. However, in spectral modeling, although the method emphasizes global structural modeling, it overlooks the fine-grained differences between local spectral bands. This limitation hinders the representation of complex spectral features and consequently restricts further improvement in overall performance.

III. Methodology

This section first introduces the overall architecture of the HSI classification framework proposed in this article. Subsequently, it provides a detailed description of three key modules improved upon MambaHSI: the DualGroupMamba module for spectral modeling, the HCA module for adaptively extracting discriminative spatial–spectral features, and the HFE module for enhancing spatial representation capability.

A. Overview

The overall architecture of our method is illustrated in Fig. 1. First, HSIs undergo preliminary feature extraction through an embedding module, which encodes the input data into a unified hidden-dimensional representation, laying the foundation for subsequent feature learning. This step is consistent with the processing approach used in MambaHSI. The resulting features are then fed into the spatial–spectral representation module (SSRM), where both spatial and spectral information are jointly extracted and fused. At this stage, the design of the DualGroupMamba module is central to perform comprehensive contextual modeling of spectral information, thereby enhancing spectral feature representation. Next, the HCA module is applied to adaptively filter interchannel attention and extract representative fusion features. This feature extraction process is repeated twice to progressively strengthen

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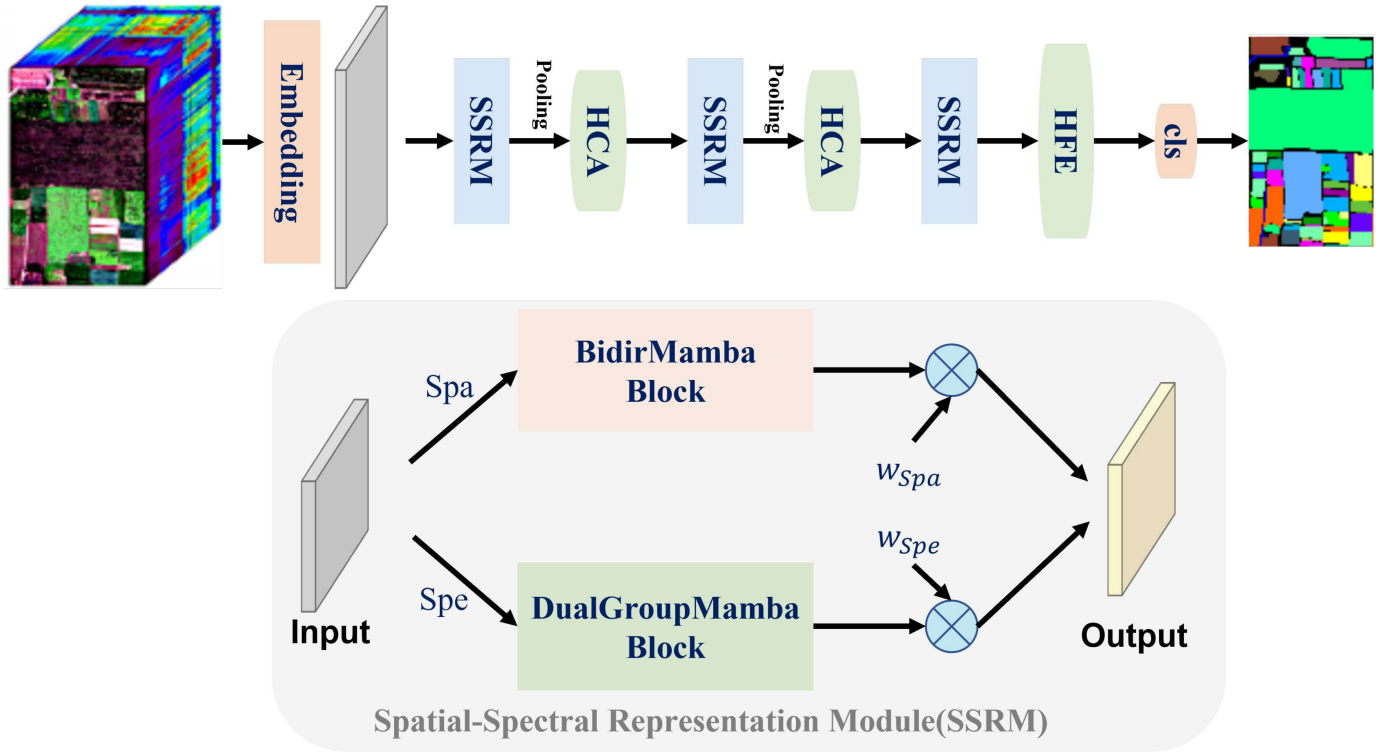


Fig. 1. Proposed framework for HSI classification, which integrates the SSRM, HCA, and HFE modules. The top part of the figure presents the overall processing pipeline. The bottom part of the figure shows the detailed structure of the SSRM, where the input features are separately processed by the spatial and spectral branches. The resulting features are then modulated by two learnable weights, w_{spa} and w_{spe}, to adaptively balance the importance of spatial and spectral representations, and are finally fused through element-wise addition.

the semantic representation. Finally, the HFE module is incorporated to further integrate local details and global spatial information, enriching the final feature representation. The fused features are then passed to the classification head for accurate prediction.

B. Spatial–Spectral Representation Module

Due to the inherent differences between spatial and spectral information in HSIs, decoupling feature extraction into two complementary branches has become a mainstream paradigm [29], [51]. Sections III-B1 and III-B2 will elaborate on the feature extraction process in detail from the perspectives of the spatial and spectral branches, respectively, whose outputs are ultimately fused via element-wise addition to form the final representation.

1) BidirMamba Block: The spatial feature extraction process defined as follows and illustrated in Fig. 2:

$$\begin{aligned}
\mathbf{F}_{\text{flat}} &= \text{flatten}\left(\mathbf{F}_{\text{embedding}}\right) \\
\mathbf{F}_{\text{flat}}^{\text{Spa}} &= \text{Mamba}\left(\mathbf{F}_{\text{flat}}\right) + \text{Reverse}\left(\text{Mamba}\left(\text{Reverse}\left(\mathbf{I}_{\text{flat}}\right)\right)\right) \\
\mathbf{F}_{\text{Spa}} &= \text{Reshape}\left(\text{GN}\left(\text{SiLU}\left(\mathbf{F}_{\text{flat}}^{\text{Spa}}\right)\right)\right) + \mathbf{F}_{\text{embedding}}. \qquad (1)
\end{aligned}$$

The input HSI is denoted as $\mathbf{F}_{\text{embedding}} \in R^{B\times H\times W\times C}$, where B, H, W, and C represent the batch size, image height, image width, and number of channels, respectively. $\mathbf{F}_{\text{flat}} \in R^{B\times L\times C}$ and $\mathbf{F}_{\text{flat}}^{\text{spa}} \in R^{B\times L\times C}$ denote the flattened input and the spatial features after bidirectional modeling, where $L = H \times W$. The Mamba refers to the standard Mamba block proposed in [52].

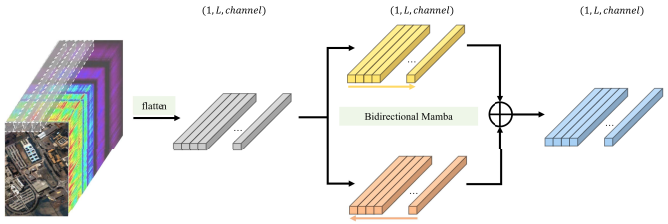


Fig. 2. Modeling process of Bidirmamba. The input is flattened along the spatial dimension and passed through Mamba in both forward and reverse directions. The two outputs are fused via element-wise addition to enhance contextual representation. The arrows indicate the direction of Mamba-based modeling.

The use of group normalization (GN) and residual connections facilitates spatial feature learning.

Unlike the spatial extraction method in MambaHSI, we adopt a bidirectional modeling strategy (BidirMamba), which updates features simultaneously in both forward and backward directions. This design enhances global contextual awareness and mitigates the incompleteness caused by unidirectional information flow. In particular, HSIs contain channels with unique and critical discriminative information; relying solely on unidirectional modeling may lead to underutilization of the spatial dependency features in some channels.

2) DualGroupMamba Block: To guide the grouping strategy in the DualGroupMamba module, we first analyze the interband correlation matrix of hyperspectral data. As shown in Fig. 3, most adjacent bands exhibit strong similarity,

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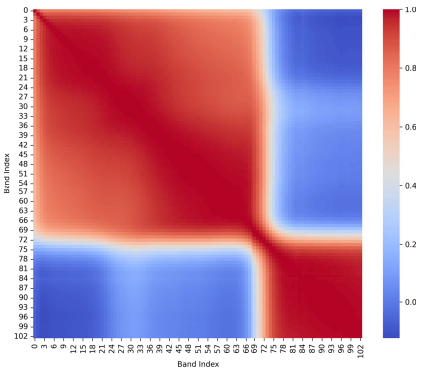


Fig. 3. Interband correlation matrix of the Pavia University dataset, illustrating the high correlation among adjacent spectral bands as the basis for locally continuous grouping.

making it difficult to identify subtle differences through direct comparison. By grouping adjacent bands, these minor variations can be accumulated and amplified, which facilitates the extraction of more discriminative features across different groups. However, if modeling is conducted solely at the intergroup level, the fine-grained variations within each group and the critical discriminative information from abrupt bands may be overlooked. To address this issue, DualGroupMamba adopts a two-stage "intragroup first, intergroup later" modeling strategy. In the local stage, it captures the subtle intragroup variations and abrupt spectral changes, while in the global stage, it models contextual dependencies across groups. This hierarchical modeling not only preserves the inherent spectral smoothness but also fully leverages the discriminative information carried by weakly correlated and abrupt bands, thereby significantly enhancing the representation ability and classification performance of spectral features.

As shown in Fig. 4, we have designed a DualGroupMamba module that also divides the spectral dimension into multiple groups. The calculation process mainly includes two stages: First (Stage 1), modeling the intragroup dependencies between spectral bands within each group; Subsequently (Stage 2), intergroup features are computed to capture global contextual information. The detailed calculation process is as follows:

$$\begin{aligned}
\mathbf{F}_{\text{group}} &= \text{SplitGroup}\left(\mathbf{F}_{\text{embedding}}\right) \\
\mathbf{F}_{\text{flat}} &= \text{flatten}\left(\mathbf{F}_{\text{group}}\right) \\
\mathbf{F}_{\text{local}} &= \text{Mamba}\left(\mathbf{F}_{\text{flat}}\right) + \text{Reverse}\left(\text{Mamba}\left(\text{Reverse}\left(\mathbf{F}_{\text{flat}}\right)\right)\right) \\
\mathbf{F}_{\text{local}}^{T} &= \text{Transpose}\left(\mathbf{F}_{\text{local}}\right) \\
\mathbf{F}_{\text{global}} &= \text{Mamba}\left(\mathbf{F}_{\text{local}}^{T}\right) + \text{Mamba}\left(\text{Reverse}\left(\mathbf{F}_{\text{local}}^{T}\right)\right) \\
\mathbf{F}_{\text{Spe}} &= \text{Reshape}\left(\text{GN}\left(\text{SiLU}\left(\mathbf{F}_{\text{global}}\right)\right)\right) + \mathbf{F}_{\text{embedding}}.
\end{aligned} \tag{2}$$

$\mathbf{F}_{\text{group}} \in R^{B\times H\times W\times G\times M}$ denotes the divided spectral group features, where G and M represent the number of groups and the number of channels per group, respectively, and $G\times M = C$. To adapt feature $\mathbf{F}_{\text{group}}$ for Mamba processing, it is transformed into structure $\mathbf{F}_{\text{flat}} \in R^{N\times G\times M}$, where $N = B \times H \times W$.

For the rearranged features $\mathbf{F}_{\text{flat}}$, we first perform intragroup modeling: each spectral subgroup (of size M) is fed into a Mamba module to capture the local dependencies among consecutive bands within the group, enabling local context modeling and information aggregation. Subsequently, during the intergroup modeling stage, the output features are rearranged via a dimension permutation operation into $\mathbf{F}_{\text{local}}^{T}$, which are then sequentially processed by another Mamba module to model long-range dependencies and global semantic relationships across different spectral subgroups. Through this two-stage "intragroup to intergroup" modeling strategy, DualGroupMamba simultaneously focuses on both local spectral details and global contextual information, significantly enhancing the model's capability to represent continuous spectral structures.

Finally, based on preserving the original embedding information, the proportion of spatial and spectral features is adaptively adjusted according to the learned weights. This allows the fused feature to retain both rich spatial–spectral information and original structural cues, thereby enhancing the performance of the final classification or recognition task

$$\begin{aligned}
\left[w_{\text{spa}}, w_{\text{spe}}\right] &= \text{Softmax}\left(\theta_1, \theta_2\right) \\
\mathbf{F}_{\text{fus}} &= w_{\text{Spa}} \times \mathbf{F}_{\text{Spa}} + w_{\text{Spe}} \times \mathbf{F}_{\text{Spe}} + \mathbf{F}_{\text{embedding}}
\end{aligned} \tag{3}$$

w_{Spa} and w_{Spe} are learnable weights for the spatial and spectral features, respectively. These weights are initialized equally and normalized via a Softmax function to ensure nonnegativity and sum-to-one constraint. Specifically, the underlying parameters θ_1 and θ_2 are initialized as trainable variables with initial values of 0.5. During training, θ_1 and θ_2 are optimized through backpropagation, enabling the model to adaptively learn the optimal weighting for each feature branch. $\mathbf{F}_{\text{Spa}}$ denotes the spatial features, and $\mathbf{F}_{\text{Spe}}$ denotes the spectral features. After performing residual connection and fusion with the original embedded features, the resulting fused feature is used as the input for subsequent computations.

C. HCA Module

The spatial–spectral information contained in HSIs is enormous, but not all channel features are helpful for classification tasks. Consistent or highly similar features are difficult to use as a basis for classification discrimination, and, therefore, can be considered redundant information or noise in classification tasks, which may further interfere with the extraction of representative features. The core idea of HCA is to guide the model to focus on feature channels with stronger discriminability, which is particularly important in HSIs.

As shown in Fig. 5(a), to enhance the discriminability of feature channels, we introduce a learnable weighted attention mechanism in the HCA module. Since hyperspectral data contain a large number of highly correlated channels along the spectral dimension, relying solely on the original features often fails to highlight discriminative information. To address this, the input features are first passed through GAP and compressed into a statistical descriptor vector for each channel, expressed as

$$z_{\text{in}} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \mathbf{F}_{\text{in}}\left(i, j\right). \tag{4}$$

The channel descriptor $z_{\text{in}} \in \mathbb{R}^{C}$ reflects the global response strength of each channel and is then fed into a channel

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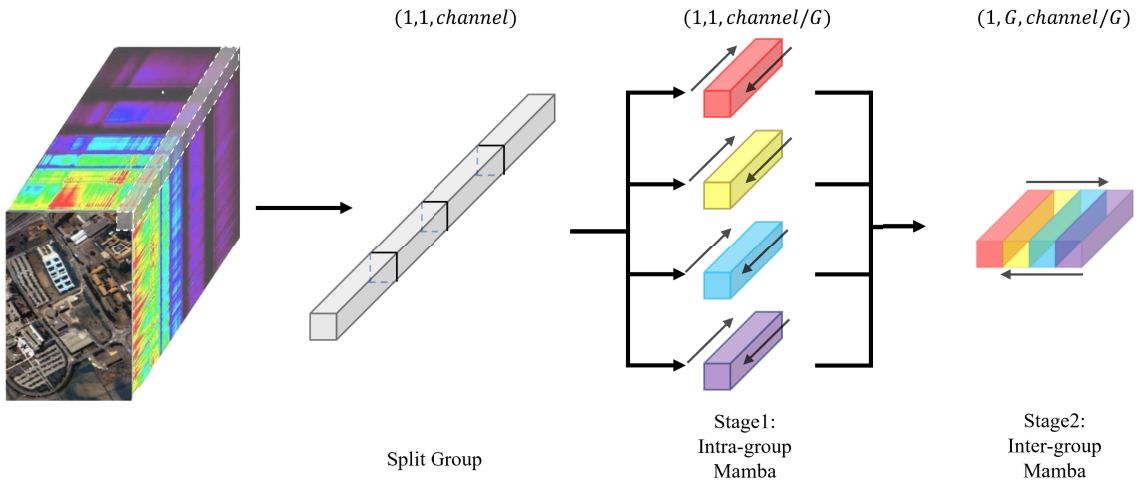


Fig. 4. Modeling process of DualGroupMamba. The spectral dimension is first divided into multiple groups, followed by a two-stage modeling process. In the first stage, intragroup dependencies among spectral bands within each group are modeled to capture local spectral patterns. In the second stage, intergroup features are computed to learn global contextual information across groups. In Stage 1 and Stage 2, the arrows indicate the modeling direction of Mamba.

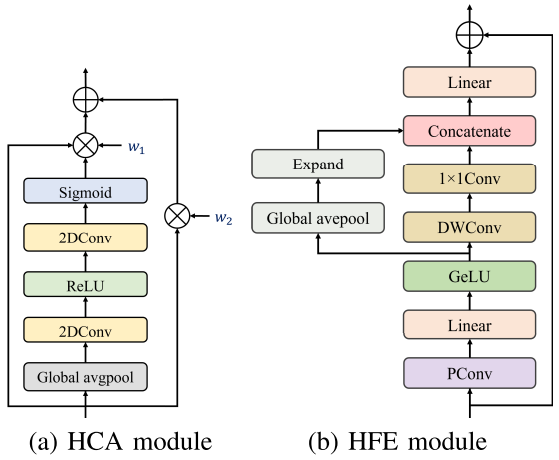


Fig. 5. Illustration of (a) HCA and (b) HFE modules.

attention network consisting of two 1×1 convolutional layers, with a ReLU activation function in between. Finally, a Sigmoid activation function is applied to produce the channel attention weights $\alpha \in \mathbf{R}^{\mathbf{C}}$

$$\alpha = \text{sigmoid}\left(\text{Conv}\left(\text{ReLU}\left(\text{Conv}\left(z_{\text{in}}\right)\right)\right)\right). \tag{5}$$

The obtained attention weights α are broadcast to each spatial location of the original feature map and multiplied with the input feature map channel-wise, thereby adaptively modeling the importance of different channels and achieving feature recalibration. To further enhance the flexibility and stability of the model, we introduce two learnable scalar weights w_1 and w_2, which are used to perform weighted fusion of the attention-enhanced features and the original features. The final output is given by

$$\mathbf{F}_{\text{out}} = w_1 \cdot (\alpha \otimes \mathbf{F}_{\text{in}}) + w_2 \cdot \mathbf{F}_{\text{in}}. \tag{6}$$

Here, both w_1 and w_2 are trainable parameters, initialized to 1, and can be automatically adjusted during training, enabling a dynamic balance between the attention mechanism and the original information. This design effectively highlights channel features that contribute to classification while preserving the original feature information, thereby enhancing the classification performance of HSIs.

D. HFE Module

The HCA module dynamically allocates channel-wise attention weights based on spatial–spectral information across different bands, primarily optimizing feature representation along the spectral dimension. Following two-stage SSRM processing and HCA-based channel filtering, we further introduce the HFE module to strengthen feature representation. The HFE module specifically targets spatial feature enhancement, leveraging both local texture details and global holistic features to improve characterization of complex spatial structures.

As shown in Fig. 5(b), the HFE module possesses both local detail and global spatial perception capabilities. Specifically, it adopts a dual-branch design: one branch employs depthwise separable convolutions to extract local information, encouraging the model to capture more detailed and discriminative texture features, while the other branch utilizes average pooling to perceive the overall structural distribution of the scene. Finally, multiscale feature fusion is achieved through channel-wise concatenation followed by linear projection. It is worth noting that although HFE also employs spatial-scale average pooling similar to HCA, it immediately upsamples the global features back to the original spatial resolution after pooling. This design not only preserves global information but also ensures spatial alignment of the feature maps. The computational steps are as follows:

$$\begin{aligned}
\mathbf{F}_{\text{mid}} &= \text{GeLU}\left(\text{Linear}\left(\text{PConv}\left(\mathbf{F}_{\text{in}}\right)\right)\right) \\
\mathbf{F}_{\text{global}} &= \text{Expand}\left(\text{GAP}\left(\mathbf{F}_{\text{mid}}\right)\right) \\
\mathbf{F}_{\text{local}} &= \text{DWConv}\left(\mathbf{F}_{\text{mid}}\right) \\
\mathbf{F}_{\text{cat}} &= \text{Concat}\left(\mathbf{F}_{\text{global}}, \mathbf{F}_{\text{local}}\right)
\end{aligned}$$

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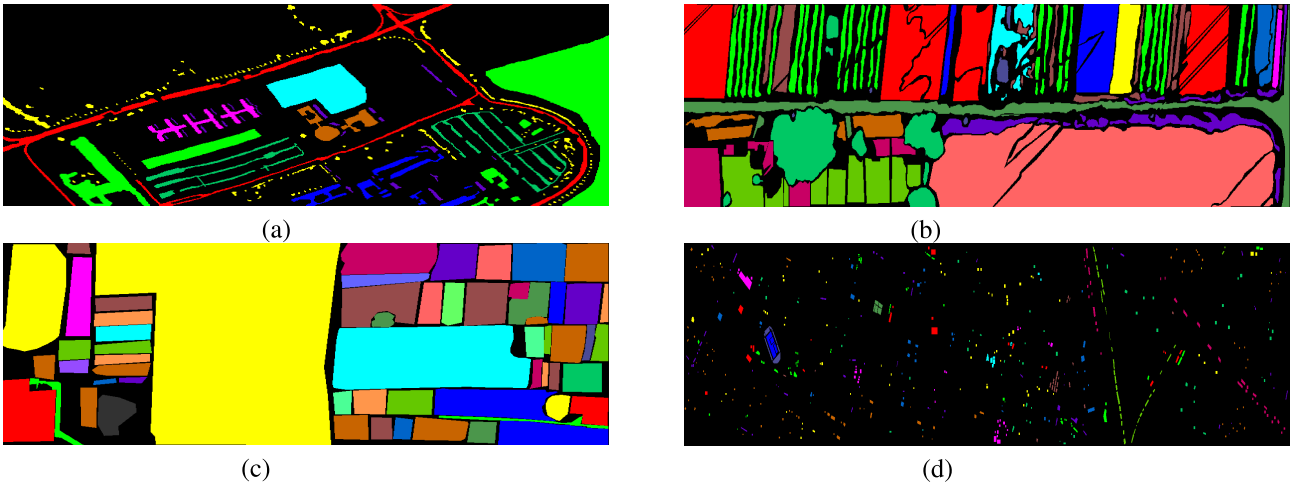


Fig. 6. Experimental HSIs: false color images across four regions: (a) Pavia University, (b) HongHu, (c) HanChuan, and (d) Houston.

$$\mathbf{F}_{\text{out}} = \text{Linear}\left(\mathbf{F}_{\text{cat}}\right) + \mathbf{F}_{\text{in}}. \tag{7}$$

The input to this module is denoted as $\mathbf{F}_{\text{in}}$ and the output as $\mathbf{F}_{\text{out}}$. By introducing the HFE module, the model not only performs key channel selection in the spectral dimension but also enables effective multiscale feature representation and fusion in the channel dimension. This significantly enhances the discriminative power and representational capacity of the final features, providing a more refined and information-rich input for subsequent classification.

E. Training and Inference

As in MambaHSI, the model takes the entire image as input during training, and a single forward pass yields predictions for all pixels in the image simultaneously [29]. The model supports end-to-end training, and the loss function $\mathcal{L}$ used during the training phase is defined as follows:

$$\mathcal{L} = \text{CrossEntropy}\left(\mathbf{I}, \mathbf{Y}_{\mathbf{tr}}\right). \tag{8}$$

Here, $\mathbf{Y}_{\mathbf{tr}}$ denotes the training labels and $\mathbf{I}$ represents the logits output by the model, both corresponding to all pixels of the entire image.

During inference, given a test image $\mathbf{I}_{\mathbf{te}}$, the final prediction $\hat{y}$ can be computed as follows:

$$\hat{y} = \text{argmax}\left(\text{softmax}\left(\text{EffMambaHSI}\left(\mathbf{I}_{\mathbf{te}}\right)\right)\right). \tag{9}$$

IV. Experiments

This section details the experimental evaluation, including the setup, comparisons with state-of-the-art methods, and an ablation study to assess module contributions.

A. Experimental Setup

1) Datasets: To thoroughly assess the effectiveness of the proposed model, we select four widely used and diverse hyperspectral datasets: Pavia University (PaviaU), Houston, WHU-Hi-HanChuan (HanChuan), and WHU-Hi-HongHu (HongHu). Refer to Fig. 6 or Table I for detailed information about the datasets.

a) Pavia University: The Pavia University dataset is an HSI dataset collected by the Reflective Optics System Imaging Spectrometer (ROSIS-3) sensor over the city of Pavia, Italy. The image comprises 610 × 340 pixels with 103 spectral bands. It is annotated with 42 776 labeled samples across nine classes, including asphalt, meadows, gravel, trees, metal sheets, bare soil, bitumen, bricks, and shadows. And the HSIs acquired by the ROSIS sensor have a spatial resolution of 1.3 m.

b) HanChuan: The WHU-Hi-HanChuan dataset was acquired in Hanchuan, Hubei province, China, with a Nano-Hyperspec imaging sensor. The study area is a rural–urban fringe zone with buildings, water, and cultivated land, which contains seven crop species: strawberry, cowpea, soybean, sorghum, water spinach, watermelon, and greens. The UAV flew at an altitude of 250 m, the size of the imagery is 1217×303 pixels. There are 274 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne hyperspectral imagery is about 0.109 m.

c) HongHu: The WHU-Hi-HongHu dataset was acquired in Honghu City, Hubei province, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor. The experimental area is a complex agricultural scene with many classes of crops, and different cultivars of the same crop are also planted in the region, including Chinese cabbage and cabbage, as well as Brassica chinensis and small Brassica chinensis. The UAV flew at an altitude of 100 m, the size of the imagery is 940×475 pixels. There are 270 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne hyperspectral imagery is about 0.043 m.

d) Houston: The Houston dataset was captured over the University of Houston and its neighboring areas using the ITRES CASI 1500 hyperspectral imager. The dataset consists of 144 spectral bands spanning from 364 to 1046 nm, with a spatial resolution of 349 × 1905 pixels. It includes 16 land-cover categories, such as residential, commercial, and vegetation types. The HSIs acquired by the hyperspectral imager have a spatial resolution of 2.5 m.

To evaluate the model, we use three widely adopted performance metrics: overall accuracy (OA), average accuracy

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TABLE I

DATASET DISTRIBUTION BY CATEGORY FOR PAVIA UNIVERSITY, HANCHUAN, HONGHU, AND HOUSTON

Pavia University			HanChuan			HongHu			Houston		
Color	Category	Test	Color	Category	Test	Color	Category	Test	Color	Category	Test
	Asphalt	6591		Strawberry	44695		Red roof	14001		Healthy Grass	1211
	Meadows	18609		Cowpea	22713		Road	3472		Stressed Grass	1214
	Gravel	2059		Soybean	10247		Bare soil	21781		Synthetic Grass	657
	Trees	3024		Sorghum	5313		Cotton	163245		Tree	1204
	Metal sheets	1305		Water spinach	1160		Cotton firewood	6178		Soil	1202
	Bare soil	4989		Watermelon	4493		Rape	44517		Water	285
	Bitumen	1290		Greens	5863		Chinese cabbage	24063		Residential	1228
	Bricks	3642		Trees	17938		Pakchoi	4014		Commercial	1204
	Shadows	907		Grass	9429		Cabbage	10779		Road	1212
				Red roof	10476		Tuber mustard	12354		Highway	1187
				Gray roof	16871		Brassica parachinensis	10975		Railway	1195
				Plastic	3639		Brassica chinensis	8914		Parking Lot 1	1193
				Bare soil	9076		Small Brassica chinensis	22467		Parking Lot 2	429
				Road	18520		Lactuce sativa	7316		Tennis Court	388
				Bright object	1096		Celtuce	962		Running Track	620
				Water	75361		Film covered lettuce	7222			
							Romaine lettuce	2970			
							Carrot	3177			
							White radish	8672			
							Garlic sprout	3446			
							Broad bean	1288			
							Tree	4000			
Total		42416	**Total**		256890	**Total**		385813	**Total**		14429

In our experiments, the training and validation sets consist of 30 and 10 samples, respectively. $N_{Total} = N_{Train} + N_{Val} + N_{Test}$.

TABLE II

QUANTITATIVE RESULT (ACC% ± STD%) OF PAVIA UNIVERSITY. BEST IN BOLD AND SECOND WITH UNDERLINE. IF TWO VALUES ARE IDENTICAL, THE ONE WITH THE SMALLER STANDARD DEVIATION (STD) IS CONSIDERED THE BETTER. THESE NOTES ARE THE SAME AS OTHERS

Class	GiGCN	CLONL	SpectraFormer	GSC-VIT	SpectralMamba	MambaHSI	S2Mamba	DBMGNet	Ours
Asphalt	90.36±1.73	**98.27±1.00**	92.60±2.29	97.10±1.64	88.46±5.24	94.35±1.73	98.10±0.23	96.69±0.38	97.23±1.10
Meadows	98.86±0.78	**98.98±0.76**	93.58±2.14	97.68±1.32	92.07±5.88	96.43±1.92	93.94±0.48	97.82±0.81	97.90±1.27
Gravel	91.66±6.33	87.83±7.44	86.46±7.42	86.46±7.42	94.07±4.71	93.39±5.35	**99.60±0.34**	96.00±1.06	97.66±2.46
Trees	81.34±10.11	97.80±1.46	92.28±6.23	91.83±9.94	92.40±2.91	89.75±3.60	**98.24±0.59**	97.56±0.48	95.49±1.28
Metal sheets	91.15±1.31	99.68±0.59	98.00±1.72	99.45±0.53	99.95±0.07	99.99±0.02	99.46±0.38	**100.00±0.00**	99.98±0.03
Bare soil	98.87±1.90	85.27±7.50	63.92±12.41	79.65±9.24	97.87±1.65	98.60±3.93	97.26±1.38	96.78±1.06	**99.35±0.36**
Bitumen	99.28±0.61	85.92±7.00	48.11±8.47	89.63±10.10	98.41±1.43	96.50±3.13	**99.81±0.12**	99.62±0.29	99.14±0.64
Bricks	87.68±7.46	90.93±4.12	81.22±4.66	89.50±1.70	96.53±2.24	94.40±2.93	96.35±0.93	94.31±1.80	**97.50±1.74**
Shadows	97.21±3.36	97.12±0.33	88.78±5.92	98.71±1.10	97.77±1.42	99.36±0.62	99.36±0.63	**99.89±0.10**	99.45±0.48
OA(%)	94.47±0.97	94.97±1.49	91.10±2.60	92.65±2.30	93.25±2.65	95.74±0.90	96.23±0.21	97.28±0.41	**97.88±0.68**
AA(%)	92.93±1.07	93.53±1.53	79.93±2.67	91.89±1.73	95.28±1.06	95.86±1.11	98.01±0.15	97.63±0.29	**98.19±0.40**
Kappa(%)	92.67±1.26	93.38±1.84	77.40±4.41	90.38±2.95	94.13±3.06	95.00±2.24	95.04±0.27	96.40±0.54	**98.12±1.51**

(AA), and Kappa coefficient (κ). All primary comparison and ablation studies are repeated ten times. The higher the values of these metrics, the better the performance of the model.

The proposed method is implemented on the PyTorch platform. To ensure stability and reliability, all experiments were conducted ten times with different random seeds. In each trial, 30 samples per category are randomly selected as the training set, ten samples are randomly chosen as the validation set, and the remaining samples are all used as the test set. The batch size for input was set to 1. The Adam optimizer was used to optimize the model parameters, with an initial learning rate of 0.0003. Here, D denotes the feature dimension of the HSI after the embedding module, which is set to 128; G represents the number of spectral groups in the DualGroup-Mamba module, which is set to 4. Our loss function calculation method is consistent with that of MambaHSI [29]. All experiments were conducted on a platform with an NVIDIA A100 GPU, a ×64 Intel Xeon Gold 6348 CPU, and 256 GB of RAM.

B. Comparison

To comprehensively evaluate the performance of the proposed method, we conducted systematic comparative experiments from both quantitative and qualitative perspectives.

The selected baselines cover diverse model architectures.

1) *CNN-Based:*
   CLONL (2024) [15], GiGCN (2022) [40].
2) *Transformer-Based:*
   GSC-ViT(2024) [47], Spectralformer(2021) [41].
3) *Mamba-Based:*
   S2Mamba (2025) [24], DBMGNet (2025) [28], MambaHSI (2024) [29], and SpectralMamba (2024) [22].

For the quantitative evaluation, we selected eight state-of-the-art HSI classification methods as benchmarks for comparison, including several Mamba-based approaches, mainly due to Mamba's notable advantages in long-range dependency modeling, computational efficiency, and model scalability. This comprehensive comparison effectively verifies the effectiveness and superiority of the proposed method.

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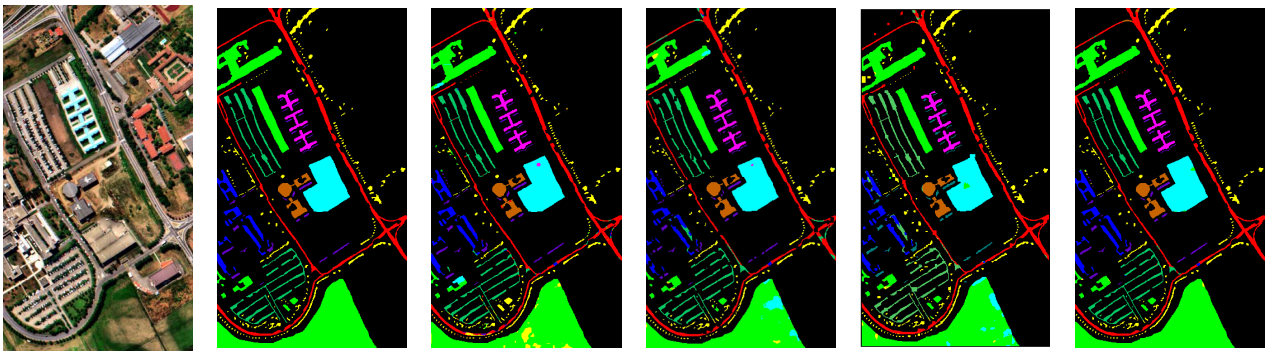


Fig. 7. Visualization results of the Pavia University dataset across various models. (a) False color image. (b) GT map. (c) MambaHSI. (d) S2Mamaba. (e) DBMGNet. (f) Ours.

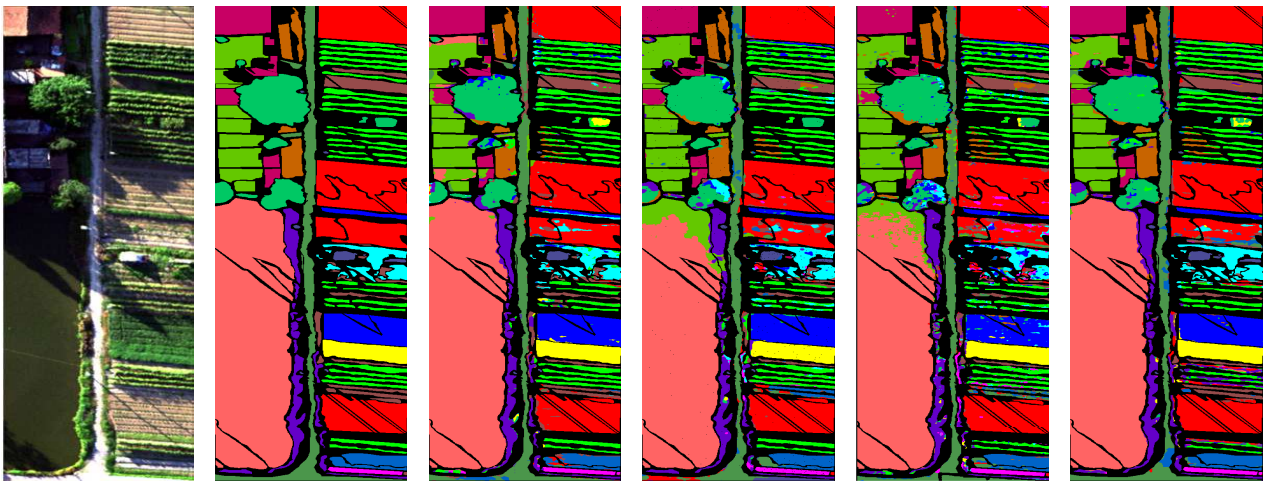


Fig. 8. Visualization results of the HanChuan dataset across various models. (a) False color image. (b) GT map. (c) MambaHSI. (d) S2Mamaba. (e) DBMGNet. (f) Ours.

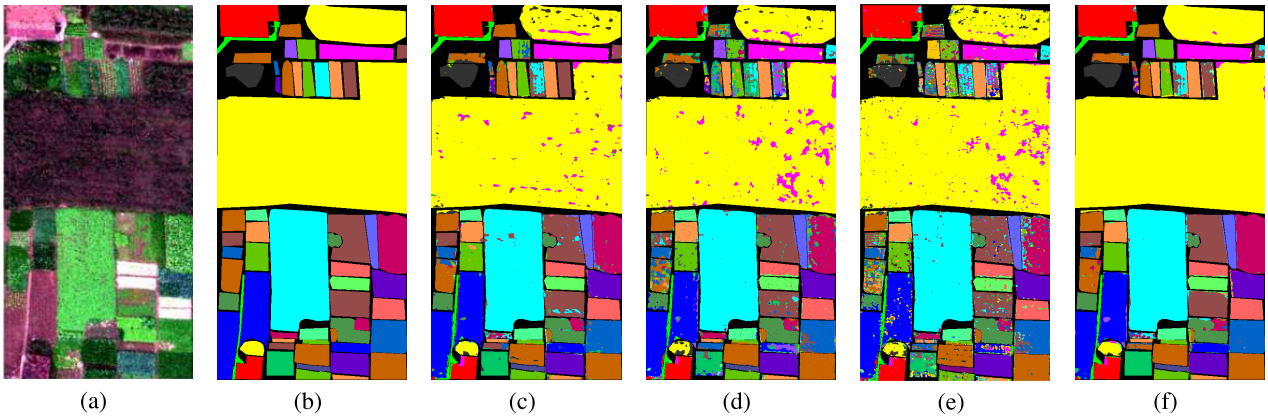


Fig. 9. Visualization results of the HongHu dataset across various models. (a) False color image. (b) GT map. (c) MambaHSI. (d) S2Mamaba. (e) DBMGNet. (f) Ours.

For the qualitative evaluation, we carefully selected three high-performing and representative recent methods for visual comparison and analysis to ensure that the evaluation results are both comprehensive and targeted.

The detailed quantitative and qualitative results are presented in Tables II–V and Figs. 7–10, respectively.

a) Pavia University: As shown in Table II, the quantitative results clearly demonstrate the superiority of our method. In terms of OA, our approach achieves 97.88%, representing a 2.14% improvement over the MambaHSI method and a 0.6% increase compared to the recently released DBMGNet. Our method also excels in AA, reaching 98.19%, which is 0.18% higher than the second-best approach. Moreover, the Kappa coefficient reaches 98.12%, indicating strong consistency between the predictions and ground-truth (GT) labels. Compared with MambaHSI (95.00%) and DBMGNet (96.40%), our approach still provides a notable improvement.

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TABLE III
QUANTITATIVE RESULT (ACC% ± STD%) OF THE HANCHUAN DATASET

Class	GiGCN	CLONL	Spectralformer	GSC-ViT	SpectralMamba	MambaHSI	S2Mamba	DBMGNet	Ours
Strawberry	89.34±2.23	**95.97±1.87**	95.13±1.25	95.26±1.50	94.50±1.79	89.72±5.75	90.82±0.45	92.45±0.46	93.96±4.29
Cowpea	81.64±3.38	**92.48±2.56**	87.27±3.37	91.22±3.05	91.52±2.41	81.37±5.69	84.25±1.21	83.83±0.64	86.51±4.40
Soybean	81.06±6.56	80.03±9.18	70.36±6.15	78.37±7.47	88.54±4.02	94.88±3.73	93.06±0.41	93.23±0.46	**95.34±3.39**
Sorghum	67.09±4.82	91.12±7.72	88.62±6.67	85.17±12.38	98.61±0.86	98.46±1.35	96.62±0.35	97.57±0.17	**98.72±1.06**
Water spinach	34.98±2.76	56.82±12.89	42.01±11.48	46.57±13.03	99.69±0.18	99.54±0.85	**99.99±0.03**	99.70±0.10	99.87±0.39
Watermelon	53.81±9.81	38.07±8.70	34.68±6.40	35.97±7.01	75.89±7.12	76.99±4.95	**92.29±0.78**	87.58±0.80	89.82±6.46
Greens	80.84±8.87	68.87±17.46	67.69±10.29	66.90±15.67	92.60±3.90	92.61±4.63	**96.29±0.31**	95.83±0.48	95.45±4.30
Trees	**89.27±4.80**	81.52±7.87	83.17±4.16	82.39±5.23	86.69±4.11	70.73±7.10	73.19±0.70	72.50±0.96	81.75±4.88
Grass	65.14±6.79	68.59±6.61	63.00±4.53	60.64±17.62	83.82±5.41	89.01±3.62	84.48±1.57	82.46±0.95	**90.50±3.85**
Red roof	93.03±4.21	90.79±14.71	88.76±12.99	93.22±4.31	95.65±2.74	94.44±2.18	**99.47±0.07**	97.10±0.35	96.44±1.61
Gray roof	93.86±5.06	87.86±4.63	77.41±13.19	82.13±7.96	93.3±4.15	96.13±1.66	**97.48±0.51**	91.91±1.22	89.57±4.32
Plastic	89.22±7.45	63.48±14.80	51.40±6.89	51.92±6.01	85.13±5.41	88.83±6.91	**97.72±0.75**	94.05±0.41	93.58±3.07
Bare soil	66.66±6.60	50.21±6.71	47.62±9.17	53.28±10.88	80.57±6.31	78.69±3.73	79.97±0.76	80.31±1.15	**81.51±3.13**
Road	91.25±2.43	90.78±2.74	88.92±3.08	90.64±3.07	91.15±2.96	89.68±3.03	81.92±2.23	74.52±0.87	**93.06±2.14**
Bright object	46.57±17.42	62.42±10.67	52.51±8.59	70.05±12.35	95.05±2.59	94.42±2.41	99.54±0.26	**99.93±0.04**	95.87±1.47
Water	99.75±0.14	98.97±1.02	99.47±0.33	**99.77±0.23**	98.81±0.75	98.28±1.62	99.52±0.08	98.66±0.17	98.59±0.80
OA(%)	86.78±1.43	85.14±2.93	82.99±2.31	83.14±3.96	91.64±1.12	90.21±1.67	91.63±0.16	90.40±0.21	**93.13±1.13**
AA(%)	76.47±1.62	76.13±3.26	71.13±2.56	73.97±2.06	90.18±1.31	89.20±1.36	91.66±0.19	90.10±0.19	**92.53±0.69**
Kappa(%)	84.64±1.62	82.75±3.31	80.29±2.61	80.57±4.33	89.99±1.63	89.07±2.31	90.24±0.19	88.81±0.25	**91.87±2.44**

TABLE IV
QUANTITATIVE RESULT (ACC% ± STD%) OF THE HONGHU DATASET

Class	GiGCN	CLONL	Spectralformer	GSC-ViT	SpectralMamba	MambaHSI	S2Mamba	DBMGNet	Ours
Red roof	90.93±2.64	95.36±5.56	96.66±1.63	**98.16±1.08**	96.37±0.62	96.47±1.50	96.03±0.39	98.10±0.04	96.73±1.01
Road	51.88±7.77	78.60±6.19	62.84±4.24	73.78±9.09	**97.40±1.32**	95.43±2.33	96.11±0.71	96.93±0.30	96.41±2.02
Bare soil	78.80±2.87	**98.22±1.32**	95.60±2.81	97.21±1.81	93.58±1.18	94.66±1.05	84.95±0.31	92.19±1.34	94.05±1.44
Cotton	99.37±0.32	99.48±0.37	99.42±0.30	**99.57±0.14**	97.82±1.65	96.96±1.79	95.43±1.10	95.12±0.45	98.97±0.60
Cotton firewood	62.41±8.43	41.33±18.92	42.14±16.25	37.03±9.33	**98.13±0.79**	96.20±1.77	96.42±0.85	96.19±0.26	97.70±1.48
Rape	89.61±3.17	96.09±1.07	95.73±1.02	95.72±1.83	94.16±1.75	93.90±2.47	92.03±0.60	**96.89±0.55**	95.81±1.98
Chinese cabbage	82.91±3.07	**92.72±1.96**	90.05±2.26	90.38±3.66	87.72±3.04	86.60±3.30	77.29±0.89	78.52±0.63	89.09±2.84
Pakchoi	75.44±8.66	46.44±6.02	35.59±5.04	32.12±4.83	97.04±2.28	96.08±2.90	79.23±0.97	85.33±0.76	**97.46±2.66**
Cabbage	85.39±3.04	97.04±2.05	97.89±1.07	97.06±2.73	96.76±1.44	94.78±1.69	**99.05±0.14**	95.87±0.34	96.91±1.42
Tuber mustard	92.42±2.80	84.88±3.47	78.81±4.58	79.23±7.53	93.35±2.25	90.45±2.72	89.62±0.99	89.38±0.59	**94.06±1.35**
Brassica parachinensis	75.17±4.50	72.03±8.35	69.29±7.09	72.41±8.95	91.82±3.30	91.96±2.87	93.75±0.67	83.20±0.85	**94.07±2.21**
Brassica chinensis	82.52±5.36	66.63±8.82	58.68±5.96	65.61±8.66	90.22±4.64	87.40±5.22	89.67±1.34	88.93±0.71	**91.00±4.20**
Small Brassica chinensis	89.28±2.70	82.67±5.83	76.20±6.07	77.49±8.47	**90.24±3.02**	89.12±3.12	87.32±1.48	71.76±1.04	90.02±2.27
Lactuca sativa	68.03±5.44	87.59±7.03	83.89±6.15	86.26±11.93	**95.58±1.75**	94.01±1.65	96.49±0.56	94.42±0.65	95.58±1.76
Celtuce	21.65±2.65	75.88±12.93	61.24±13.37	63.62±20.25	99.69±0.73	98.84±1.17	**99.81±0.12**	99.24±0.15	99.15±0.80
Film covered lettuce	71.70±6.79	97.39±1.73	95.23±2.99	97.42±1.66	97.35±1.76	96.83±1.60	**99.49±0.30**	97.16±0.20	97.42±1.85
Romaine lettuce	54.94±4.04	80.09±9.70	68.68±8.79	84.92±6.09	**99.10±1.25**	98.60±1.49	98.05±0.69	96.24±0.23	98.65±1.75
Carrot	67.65±8.17	65.55±15.89	60.99±7.01	67.44±10.50	98.47±0.98	96.72±1.82	98.78±0.25	**99.50±0.22**	98.43±0.99
White radish	72.71±8.86	82.53±7.23	82.77±4.78	85.10±8.11	92.89±2.15	91.30±3.07	94.63±0.34	**94.92±0.30**	93.68±2.61
Garlic sprout	66.48±6.47	81.38±6.60	71.18±8.29	71.64±10.46	**99.42±0.63**	99.18±0.42	96.60±0.60	98.95±0.19	99.25±0.48
Broad bean	65.20±14.58	30.19±12.06	28.91±8.18	25.92±7.91	**100.00±0.00**	99.91±0.18	99.98±0.05	99.01±0.44	99.94±0.19
Tree	85.35±8.84	55.02±10.89	58.01±15.28	46.26±11.15	99.64±0.37	99.54±0.51	99.38±0.21	95.36±0.34	**99.86±0.34**
OA(%)	87.22±0.60	87.93±2.11	86.04±2.83	86.18±1.86	95.41±1.07	94.58±1.01	92.65±0.53	92.28±0.31	**96.31±0.39**
AA(%)	74.08±1.50	77.60±2.25	73.17±2.45	74.74±1.64	95.76±0.61	94.77±0.61	93.64±0.19	92.87±0.16	**96.10±0.36**
Kappa(%)	83.95±0.73	85.06±2.49	82.72±3.22	82.93±2.16	95.05±1.57	94.00±1.88	90.82±0.64	90.35±0.38	**95.76±1.30**

The visual results further validate the advantages of our method. As shown in Fig. 7, most existing models exhibit significant misclassification over three large meadow regions in the image. In contrast, our approach accurately identifies these areas. This result further confirms that comprehensive contextual modeling and efficient feature extraction play a critical role in improving HSI classification accuracy.

b) HanChuan: As shown in Table III, our method demonstrates significant performance advantages on the HanChuan dataset, achieving an OA of 93.13%, an AA of 92.53%, and a Kappa coefficient of 91.87%. Compared with MambaHSI, our method improves OA, AA, and Kappa by 2.92%, 3.33%, and 2.8%, respectively, thereby enhancing both classification accuracy and model stability. Our method surpasses all other methods in AA and Kappa, also exhibiting comprehensive and reliable classification capabilities. These results indicate that our approach effectively models both spectral and spatial features while maintaining a strong balance between accuracy and robustness, making it especially suitable for real-world hyperspectral scenarios like HanChuan, which exhibit complex spatial and spectral characteristics.

The visual comparison further confirms the superiority of our method. As shown in Fig. 8, most existing models make substantial classification errors in large water areas affected by shadows, whereas our method achieves more accurate recognition and classification across the entire water region. Additionally, in mixed regions such as trees, gray rooftops, and grasslands, our method demonstrates stronger boundary discrimination, yielding higher classification accuracy and fewer misclassifications. These visual improvements are consistent

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TABLE V
QUANTITATIVE RESULT (ACC% ± STD%) OF THE HOUSTON DATASET

Class	GiGCN	CLONL	Spectralformer	GSC-ViT	SpectralMamba	MambaHSI	S2Mamba	DBMGNet	Ours
Healthy Grass	83.21±5.03	88.24±9.00	91.29±5.38	90.16±6.48	93.72±5.13	**95.20±4.66**	93.28±0.05	94.67±1.94	94.10±4.09
Stressed Grass	82.83±7.41	93.41±5.99	92.12±6.99	95.77±2.95	95.68±1.86	**98.29±1.26**	97.35±2.71	97.45±1.46	97.12±1.53
Synthetic Grass	98.83±1.66	95.21±9.84	94.68±3.25	99.83±0.26	99.73±0.35	99.74±0.46	99.73±0.11	**99.86±0.20**	99.77±0.42
Tree	72.42±10.94	97.44±3.80	93.32±9.22	94.78±7.69	96.51±2.05	97.26±2.47	**98.85±0.24**	98.26±1.71	97.29±2.21
Soil	93.14±4.62	97.07±2.86	98.18±1.72	97.63±2.69	99.82±0.24	99.54±0.76	**100.00±0.00**	**100.00±0.00**	99.39±1.63
Water	88.99±6.78	94.85±4.72	87.21±8.11	96.03±2.59	96.95±2.94	97.47±2.33	92.07±0.79	**100.00±0.00**	98.56±1.72
Residential	82.34±5.24	94.02±2.94	91.47±3.54	93.11±2.39	93.46±2.97	93.03±2.25	92.00±1.57	89.27±2.16	**94.61±2.63**
Commercial	91.02±6.25	**93.80±3.94**	84.41±7.91	90.53±8.57	79.22±4.15	81.35±3.98	81.87±1.20	86.54±2.42	83.48±3.57
Road	85.51±6.43	88.32±2.83	87.75±3.83	85.20±4.96	91.19±3.18	90.03±3.40	89.86±0.56	87.05±2.58	**92.63±2.72**
Highway	94.92±2.69	76.55±6.96	82.09±5.12	85.28±4.22	97.63±2.07	96.55±1.49	94.13±0.88	96.88±2.86	**98.77±1.38**
Railway	**96.52±3.78**	91.57±4.40	85.90±4.63	93.08±3.33	93.85±2.66	92.74±2.18	93.40±4.64	94.31±3.52	95.84±1.25
Parking Lot 1	92.45±3.63	83.37±6.17	85.33±5.87	88.46±3.14	92.25±3.69	91.19±3.68	95.38±0.52	**95.64±3.26**	93.21±2.84
Parking Lot 2	88.10±5.13	89.11±4.64	86.81±6.28	90.01±4.69	96.34±2.47	97.67±2.04	**98.30±1.55**	95.50±0.15	97.09±2.54
Tennis Court	98.30±2.67	97.98±2.63	91.65±6.36	97.01±5.11	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.79±0.15	99.97±0.08
Running Track	99.14±1.89	96.08±3.96	96.18±2.66	97.73±1.23	96.76±1.29	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	96.45±2.63
OA(%)	88.13±2.26	90.31±1.73	89.13±2.12	91.85±1.05	94.11±0.80	94.46±0.83	94.44±0.47	94.83±0.87	**95.24±0.70**
AA(%)	89.85±1.91	91.80±1.46	89.89±1.74	92.97±0.92	94.87±0.79	95.34±0.78	95.08±0.42	95.68±0.69	**95.89±0.65**
Kappa(%)	87.16±2.44	89.52±1.87	88.24±2.29	91.18±1.14	94.00±0.80	94.21±1.93	93.99±0.51	94.41±0.94	**95.36±1.79**

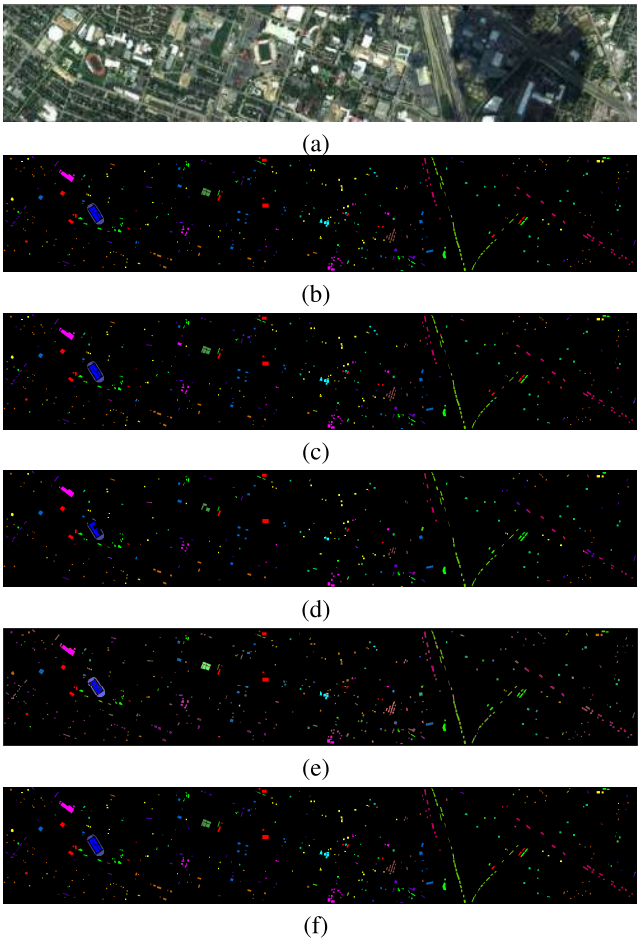


Fig. 10. Visualization results of the Houston dataset across various models. (a) False color image. (b) GT map. (c) MambaHSI. (d) S2Mamaba. (e) DBMGNet. (f) Ours.

with the higher Kappa and AA scores, which can be attributed to the use of the HCA module in our method to extract more discriminative features, thereby enabling precise classification in complex areas.

c) HongHu: As shown in Table IV, our method demonstrates groundbreaking classification performance on the HongHu dataset, achieving an OA of 96.31%, an AA of 96.10%, and a Kappa coefficient of 95.76%, significantly outperforming existing mainstream methods. Compared to MambaHSI, our method improves OA, AA, and Kappa by 1.73%, 1.33%, and 1.76%, respectively, further underscoring its superior performance. This outstanding result is largely attributed to our proposed spectral selection strategy, which effectively suppresses redundant information while enhancing the discriminability of feature representations, achieving not only high classification accuracy but also excellent stability—ranking in the top 2 across 10 out of 22 categories.

The visual results highlight our method's unique advantages in key classification scenarios. As shown in Fig. 9, when classifying large cotton areas, our method almost perfectly reconstructs the GT, demonstrating exceptional accuracy and boundary completeness. Notably, the HongHu dataset contains many types of densely distributed and visually similar crops, such as rape and Chinese cabbage, which substantially increase the complexity of the classification task. Despite this, our method is still able to accurately distinguish the majority of target categories outside these small and difficult regions, whereas other methods tend to suffer from confusion and misclassification, significantly impacting their overall classification performance. Similar to the HanChuan dataset, classification in complex regions relies more heavily on the feature representation capability. Our method can effectively extract highly discriminative features, thus demonstrating stronger adaptability and robustness in complex scenarios.

d) Houston: As shown in Table V, our method also achieves the best classification performance on the Houston dataset, with an OA of 95.24%, an AA of 95.89%, and a Kappa coefficient of 95.36%, all of which significantly outperform existing mainstream methods. Compared to the current second-best method, DBMGNet, our approach improves OA, AA, and Kappa by 0.41%, 0.21%, and 0.95%, respectively. This performance advantage is largely attributed to our introduction of cross-scale spatial feature perception and modeling strategies into the network, which effectively enhances the model's ability to understand complex spatial structures.

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TABLE VI
COMPARATIVE PERFORMANCE ON FOUR HSI DATASETS

Method	GiGCN	CLONL	Spectralformer	GSC-VIT	SpectralMamba	MambaHSI	S2Mamba	DBMGNet	Ours
OA(%)	89.15±1.31	89.59±2.06	87.31±2.46	88.46±2.29	93.60±1.41	93.75±1.10	93.74±0.34	93.70±0.45	**95.64±0.73**
AA(%)	83.33±1.53	84.77±2.13	78.53±2.36	83.39±1.59	94.02±0.94	93.79±0.97	94.60±0.24	94.07±0.33	**95.68±0.53**
Kappa(%)	87.11±1.51	87.68±2.38	82.16±3.13	86.27±2.65	93.29±1.77	93.07±2.09	92.52±0.40	92.49±0.53	**95.28±1.76**

TABLE VII
ABLATION STUDY OF KEY COMPONENTS OF OUR METHOD

Base	BidirMamba	DualGroupMamba	HCA	HFE	PaviaU	HanChuan	HongHu	Houston
✓					95.76±0.91	90.19±1.52	94.57±1.29	93.93±0.84
✓	✓				97.06±0.62	91.27±1.71	94.90±0.81	94.92±0.66
✓	✓	✓			97.28±0.54	91.29±1.22	94.97±0.96	94.31±0.98
✓	✓	✓	✓		97.19±0.85	92.05±0.74	95.31±1.32	94.36±0.64
✓	✓	✓		✓	96.98±2.32	92.34±1.02	95.85±0.55	95.05±1.04
✓	✓		✓	✓	97.84±0.81	92.87±1.10	96.19±0.41	95.20±0.76
✓	✓	✓	✓	✓	**97.88±0.68**	**93.13±1.13**	**96.31±0.39**	**95.24±0.70**

The visual comparison results shown in Fig. 10 further validate the effectiveness of the proposed strategy. In the Houston dataset, the "grass" category includes three different types, which are often confused during classification due to similar spectral and textural features—posing a significant challenge in real-world scenarios. From the right side of the image, it is clearly observable that after comprehensive cross-scale modeling, our method distinguishes between different types of grass regions more accurately than other models, significantly reducing interclass confusion and further improving the overall classification accuracy and robustness. In datasets like Houston, which feature large spatial scales and complex details, the model relies more on the comprehensive capability of the HFE module to handle wide-area variations and preserve fine-grained details during feature extraction, thereby enhancing overall classification performance.

Table VI presents the average overall classification accuracy of various methods across the four datasets. Based on repeated experiments with ten different random seeds, our method achieved the highest mean OA of 95.64%, surpassing the current best-performing method by 1.89 percentage points. Although the recently proposed DBMGNet shows slightly better result stability (σ= ±0.45) compared to our method (σ = ±0.73), the significant advantage in classification accuracy clearly demonstrates that our method effectively enhances the model's ability to represent spatial–spectral features in HSIs.

C. Ablation Study

To thoroughly assess the individual contributions of different architectural components, we perform a series of ablation experiments across four benchmark datasets: Pavia University, HongHu, HanChuan, and Houston. Table VII presents the ablation study results based on OA, where checkmarks (✓) indicate the model performance with corresponding components retained.

The baseline model (Base) serves as a reference without any additional modules. After incorporating the BidirMamba module, classification performance improved across all datasets, with particularly significant gains observed on the HanChuan and PaviaU datasets. This indicates that BidirMamba possesses stronger contextual modeling capabilities in the spatial domain, enabling more effective capture of spatial structural features in images and thus significantly enhancing feature representation. All experimental results strongly support the conclusion that HSIs, characterized by large spatial scales and complex structures, require comprehensive modeling of spatial contextual information to achieve high classification performance.

With the addition of the DualGroupMamba module, the classification performance improved most significantly on the PaviaU dataset, while only slight improvements were observed on the HanChuan and HongHu datasets, and performance even declined on the Houston dataset. This is primarily because, although DualGroupMamba enables more comprehensive context modeling, it lacks mechanisms to suppress spectral redundancy and strategies for multiscale spatial modeling. As a result, the added contextual information is difficult to convert into effective discriminative features in complex scenarios. This issue is especially evident in the Houston dataset, which contains the most complex spectral and spatial information, where excessive contextual input leads to redundancy and degrades performance. This phenomenon further validates our argument: relying solely on context modeling, while neglecting spectral feature selection and effective spatial structure modeling, fails to deliver substantial improvements in classification performance. In contrast, for datasets like PaviaU, which have fewer spectral bands and relatively simple structures, context modeling alone can yield noticeable performance gains. For datasets with higher spectral dimensionality and more complex spatial structures (such as HanChuan, HongHu, and Houston), the further introduction of spectral feature selection and spatial modeling optimization mechanisms becomes particularly crucial.

To rigorously assess the contribution of each module, we conducted ablation experiments by separately adding HCA and HFE to the base framework equipped with BidirMamba and DualGroupMamba. This design allows us to isolate the effects of spectral selection and spatial enhancement in a controlled manner. With the HCA module alone, the classification performance on the HanChuan and HongHu datasets improved

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significantly, demonstrating the importance of selecting discriminative spectral and spatial features when dealing with spectrally complex data. In contrast, the performance gain on the Houston dataset was relatively limited, mainly because its large spatial scale relies more on effective extraction and fusion of multiscale spatial features. On the PaviaU dataset, however, the addition of the HCA module led to a performance decline. This may be attributed to its limited number of spectral bands, where further feature selection weakened the overall feature representation. Therefore, for such datasets, performance relies more on thorough context modeling rather than feature selection.

After adding the HFE module alone, classification performance significantly improved across most datasets, indicating that the module effectively enhances the model's ability to perceive and model cross-scale spatial features—particularly beneficial for datasets with complex spatial structures and large-scale spatial distributions. However, for datasets like PaviaU, which have smaller spatial scales and simpler texture structures, overly aggressive spatial modeling may introduce redundant information or noise, thereby slightly impairing classification performance.

From the final experimental results, the FULL model demonstrates outstanding performance. This indicates that after sufficient contextual modeling in both spectral and spatial domains—combined with effective feature selection as well as multiscale feature extraction and integration—the model can achieve optimal results. The ablation study further validates the effectiveness of each module and shows that their interaction leads to the best overall performance. To gain a clearer understanding of the contribution of each module, we further analyze the independent ablation of HCA and HFE.

Specifically, compared to the case without HCA, the FULL model adaptively highlights the most discriminative spectral bands while suppressing redundant information, thereby achieving significant classification improvements on the Pavia University, HanChuan, and HongHu datasets. However, on the Houston dataset, the performance gain of HCA is relatively limited. This is mainly due to its larger spatial scale and more complex class distribution, where model performance relies more heavily on effective multiscale spatial context modeling and cross-region feature fusion.

On the other hand, compared to the case without HFE, when spectral discriminability has already been significantly enhanced by HCA and the previously obscured information has been fully exploited, the addition of HFE further strengthens spatial representation. In this way, spatial enhancement complements the high-quality spectral features provided by HCA, leading to consistent performance improvements across all datasets. This further demonstrates that both HCA and HFE contribute effectively when used independently, while their combination produces complementary effects, ultimately driving the FULL model to achieve optimal performance.

To further evaluate the role of DualGroupMamba within the complete framework, we conducted additional ablation experiments comparing the full models that incorporate either the Spectral Mamba or DualGroupMamba in the base. The results show that, with the integration of HCA and HFE modules, DualGroupMamba brings performance improvements across all datasets, with particularly notable gains on spectrally complex datasets such as HanChuan and HongHu.

TABLE VIII
CLASSIFICATION RESULTS UNDER DIFFERENT GROUP SETTINGS

Groups (G)	PaviaU	HanChuan	Honghu	Houston
2	97.60±1.37	92.32±1.08	96.16±0.39	95.08±0.85
4(Ours)	**97.88±0.68**	**93.13±1.13**	**96.31±0.39**	**95.24±0.70**
8	97.51±1.04	92.46±1.21	96.18±0.37	94.77±1.32
16	97.61±0.81	92.81±1.25	96.15±0.30	95.03±0.93
32	97.83±0.68	92.84±1.37	96.31±0.30	95.04±0.60
64	97.21±2.28	91.75±2.28	96.20±0.54	95.16±0.96

TABLE IX
COMPARISON OF HFE WITH OTHER SPATIAL ENHANCEMENT MODELS

Models	PaviaU	HanChuan	Honghu	Houston
FRFN	97.28±0.89	90.91±1.53	94.58±1.07	94.45±0.94
SGE	97.42±0.59	92.11±1.69	95.53±0.58	94.64±0.94
RFCBAMConv	95.75±0.97	86.13±1.77	89.19±2.33	93.84±0.80
HFE(Ours)	**97.88±0.68**	**93.13±1.13**	**96.31±0.39**	**95.24±0.70**

Building upon these findings, we further investigated the adaptability of DualGroupMamba to different group settings (G), aiming to better understand how the intra- and intergroup modeling strategy affects performance. We conducted comparative experiments with different G values (G = 2, 4, 8, 16, 32, 64) across multiple datasets, as shown in Table VIII. The results show that the model achieves the best overall performance when G = 4. Notably, G = 4 provides a good balance between intragroup and intergroup modeling: too few groups lead to excessively large intragroup ranges, which may obscure subtle differences, while too many groups fragment intergroup information, making it difficult to capture global dependencies effectively. Although performance fluctuates across different values of G, the overall variation remains relatively small, indicating that the model maintains strong performance over a wide range of group settings and exhibits a certain degree of hyperparameter robustness.

To demonstrate the innovation of the proposed HFE module, we conducted a comparative analysis with several existing spatial feature enhancement methods, such as FRFN [53], SGE [54], and RFCBAMConv [55], as shown in Table IX. Although these methods emphasize either local or global modeling, they commonly suffer from issues such as insufficient information utilization, imprecise channel selection, or a lack of spectral modeling capability. In contrast, HFE integrates deep convolution and global pooling, and introduces partial channel convolution to effectively highlight key features and suppress redundant interference. Additionally, it incorporates 1-D convolution to preserve spectral information. These design choices result in superior enhancement performance and adaptability, showcasing the rationality and practicality of the HFE structure.

To highlight the uniqueness of the proposed HCA module in HSI processing, we conduct a comparative analysis with typical channel attention mechanisms such as SE [56] and ECA [57], as shown in Table X. SE compresses features via GAP and fully connected layers, while ECA captures local dependencies across channels using 1-D

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TABLE X
COMPARISON OF CHANNEL ATTENTION MECHANISMS

Models	PaviaU	HanChuan	Honghu	Houston
SE	97.75±0.79	90.72±1.16	96.03±0.54	95.24±0.90
ECA	97.30±1.25	89.66±2.15	96.22±0.51	95.01±0.65
HCA(Ours)	**97.88±0.68**	**93.13±1.13**	**96.31±0.39**	**95.24±0.70**

TABLE XI
COMPARISON OF MODEL EFFICIENCY IN TERMS OF PARAMETER COUNT, GFLOPS, INFERENCE TIME PER IMAGE (MEASURED ON THE PAVIA UNIVERSITY DATASET)

Model	Params(M)	GFLOPs	Inference Time(s)	OA	AA	Kappa
GSC-ViT	0.15	2026.30	49.44	92.65	91.89	90.38
SpectralFormer	0.17	9086.19	62.43	91.10	79.93	77.40
S2Mamba	**0.13**	48.26	10.59	97.01	96.95	95.99
DBMNet	1.37	99.70	49.75	95.78	96.26	94.34
MambaHSI	0.41	**39.18**	**0.03**	95.74	95.86	95.00
ours	0.85	101.72	0.94	**97.88**	**98.19**	**98.12**

convolution. In contrast, HCA focuses on selectively enhancing spatial–spectral features that are critical for classification along the channel dimension. It models interchannel interactions using a lightweight convolutional structure and adaptively balances original and enhanced features through a dual-branch residual fusion strategy. This effectively improves feature discriminability and suppresses redundant noise. The design is particularly well-suited for the complex spatial–spectral coupling in hyperspectral data, demonstrating stronger classification capability in multiclass recognition tasks.

D. Model Efficiency Analysis

Given the large-scale and high dimensionality of hyperspectral data, computational efficiency becomes a critical factor in real-world applications. To further demonstrate the practicality of our proposed method, we conduct a comprehensive comparison in terms of model complexity and inference time against several representative approaches. This analysis highlights the lightweight nature and fast inference speed of our model, making it well-suited for deployment in scenarios with limited computational resources or real-time requirements.

As shown in Table XI, our method achieves the highest OA, AA, and Kappa coefficient among all compared approaches. Although the number of parameters increases from 0.41M to 0.85M and the inference time rises from 0.03 to 0.94 s compared to our baseline model MambaHSI, our method significantly reduces the number of floating-point operations (GFLOPs) compared to mainstream Transformer-based models such as GSC-ViT and SpectralFormer, demonstrating a clear computational advantage. Even when compared with other Mamba-based methods (e.g., S2Mamba and DBMNet), although our model may not have the lowest parameter count or GFLOPs, it consistently achieves superior inference speed and classification performance.

Regarding the additional computational cost introduced by the bidirectional propagation mechanism in our model, we provide the following clarification. Compared with the original MambaHSI architecture, our method incorporates a more refined hierarchical two-stage spectral modeling strategy and further enhances bidirectional modeling in the spatial domain. These enhancements inevitably increase the computational overhead and parameter count, but they also lead to substantial improvements in classification accuracy, making the additional resource consumption acceptable in practice.

In summary, our method achieves a favorable tradeoff between model complexity and computational efficiency. By leveraging the linear computational characteristics of the Mamba architecture, our model maintains high efficiency while delivering significant performance gains, successfully balancing accuracy and efficiency.

V. CONCLUSION

In this article, we propose an enhanced HSI classification framework that significantly boosts accuracy through three key components: 1) the DualGroupMamba module employs a two-stage spectral modeling strategy to capture both local and global spectral dependencies; 2) the HCA module utilizes channel attention to adaptively select spatial–spectral features, enhancing discriminability; and 3) the HFE module integrates depthwise convolution and global pooling to strengthen multiscale spatial modeling. Through a comprehensive evaluation on four benchmark hyperspectral datasets, the proposed method achieves an average overall classification accuracy of 95.64%, outperforming the second-best baseline method (MambaHSI) by 1.89%. Future work will explore adaptive spatial modeling, dynamic feature selection, cross-modal fusion (e.g., LiDAR or SAR), lightweight network design for efficiency, and generalization-aware architectures for broader applicability.

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